The Standard Model from One Transvection

The discrete Standard-Model anatomy as the geometry of a single long-root transvection on the symplectic quadrangle $W(3,3)$

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Abstract

We show that the entire *discrete* structure of the Standard Model — the gauge group $SU(3)\times SU(2)\times U(1)$ with its color/electroweak factorization, exactly three fermion generations, the discrete flavor group S_3, the Yukawa selection rule, chirality, parity, charge-conjugation, the fermion content of one generation, the gauge connection and its curvature, and a gravitational partition function — is the representation-theoretic geometry of a single *long-root transvection* R on the symplectic generalized quadrangle $W(3,3)$ (the 40 totally isotropic points of $\mathbb{F}_3^4$, with $\mathrm{Aut} = \mathrm{PSp}(4,\mathbb{F}_3)$ of order 25920). The gauge group is the *centralizer* $C(R)$; the three generations are its *center*; the flavor symmetry is its *normalizer*; parity is the W/Q duality; the Yukawa texture is the $\mathbb{Z}_3$ grading R induces on the matter shell; and the gauge curvature is the *commutator* of two generation centers, an $\mathfrak{su}(2)$-valued field strength supported on the matter graph. Every statement is a theorem with an executable witness (the BT858–922 chain); there are zero free parameters. The mixing angles are fixed in structure to the single scale Φ_3 (the PMNS is tribimaximal $\pm 1/\Phi_3$; the Cabibbo angle is the q/Φ_3 rotation), matching all nine flavor observables within 1σ bar one. The remaining physics — the coupling magnitudes, the masses, and the curved-4D continuum limit — is the established numerical/analytic layer and is not claimed here.

1 The datum

The symplectic quadrangle $W(3,3)$ has 40 points (the totally isotropic 1-spaces of $\mathbb{F}_3^4$ under a non-degenerate alternating form), with collinearity graph the strongly regular $\mathrm{SRG}(40,12,2,4)$ and automorphism group $\mathrm{PSp}(4,\mathbb{F}_3) \cong W(E_6)^+$ of order 25920. A *long-root transvection* R (an order-3 element fixing a hyperplane pointwise) is the single datum from which the discrete Standard Model unfolds. We fix a point p_0; R is the center of the Heisenberg radical of $\mathrm{Stab}(p_0)$, fixing the 13-point perp-plane $p_0^\perp$ and acting freely (nine orbits of three) on the 27 non-collinear points (the *matter shell*).

2 The matter sector

The integral homology of the $W(3,3)$ line-triangle 2-complex (40 vertices, 240 edges, 160 in-line triangles) is $H_0 = \mathbf{1}$, $H_1 = \mathbf{81}$, $H_2 = \mathbf{40}$.

Theorem 1 (Steinberg register, BT861). *$H_1 \otimes \mathbb{C}$ is irreducible of dimension $81 = q^4$: it is the* Steinberg module *of* $\mathrm{PSp}(4,\mathbb{F}_3)$ *($\langle \chi,\chi\rangle = 1$, verified over all 25920 elements, with $\chi(g) = \#\mathrm{flags} - \#\mathrm{pts} - \#\mathrm{lines} + 1$). It equals the logical space of the $[[240,81,4,3]]_3$ CSS code: matter memory is Schur-protected.*

Theorem 2 (Three generations from Steinberg vanishing, BT863)**.** $\chi_{\mathrm{St}}(g) = 0$ *iff* $3 \mid \mathrm{ord}(g)$. *Hence* any *order-*3 *symmetry splits the* 81*-dimensional matter register into* $27+27+27$*: the three generations are forced, not chosen. The matter shell* $\mathbb{C}[27]$ *likewise grades* $9+9+9$ *under* R*.*

The 27 matter shell is a torsor under the Heisenberg group $N = 3^{1+2} = O_3(\mathrm{Stab}(p_0))$ (BT858), and R is the long-root transvection realizing Pillar 68's Yukawa triality R (nine free orbits on the shell, BT874).

3 The gauge sector

Theorem 3 (Gauge group, BT876/887)**.** *The centralizer* $C(R) = \mathrm{Stab}(p_0)$ *(order* 648*) acts on the* $12 = k$ *gauge bosons with permutation rank* 3*, giving the module*

$$\mathbb{C}[12] = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \oplus SU(2) \oplus SU(3),$$

the Standard-Model gauge group. It factors as $C(R) = 3^{1+2} : SL(2,3) = 27{:}24$*: the normal Heisenberg radical (order* $q^q = 27$*,* color*) fixes the electroweak* $\mathbf{1} \oplus \mathbf{3}$ *pointwise (the* $W/Z/\gamma$ *are color-blind) and acts on the gluon octet* $\mathbf{8}$*; the Levi* $SL(2,3)$ *(order* $f = 24$*,* electroweak*) carries the* $\mathbf{1} \oplus \mathbf{3}$*.*

Corollary 4 (Generations are the gauge center, BT880)**.** $Z(C(R)) = \langle R \rangle \cong \mathbb{Z}_3$*: the three generations are the center of the gauge group, acting trivially on the adjoint exactly as the* $\mathbb{Z}_3$ *center of color* $SU(3)$ *acts on the gluon octet. "Generations are gauge-blind" because the generation symmetry* is *the gauge center.*

Corollary 5 (Color is the matter Heisenberg group, BT888)**.** *The color radical* N *is the* same *Heisenberg group that acts simply transitively on the matter shell. Matter carries color because the* 27*-shell is a torsor under the color group: color rotations are matter-shell motions.*

Theorem 6 (Fermion content of one generation, BT889)**.** $C(R)$ *is transitive on the matter shell with point-stabilizer the electroweak Levi* $SL(2,3)$*, so* $27 = C(R)/SL(2,3)$*; rank* 6*, suborbits* $[1,1,1,8,8,8]$*, module* $\mathbf{1} \oplus \mathbf{6} \oplus \mathbf{8} \oplus \mathbf{12}$*. The three electroweak-fixed points are color-singlet (lepton-like); the three color-*8 *orbits are colored (quark-like):* $27 = 3 + 24$*.*

4 The discrete flavor/parity structure

Theorem 7 (Flavor S_3 and CP, BT878/879)**.** *The normalizer datum* $N(\langle R \rangle)/C(R) = \mathbb{Z}_2$ *supplies a charge-conjugation* C *(*$CRC^{-1} = R^{-1}$*) inverting the generation grade;* $\langle R, C \rangle = S_3$ *is the flavor group, under which the matter shell decomposes* $6 \cdot \mathbf{1} \oplus 3 \cdot \mathbf{1}' \oplus 9 \cdot \mathbf{2}$ *(the standard doublet appearing* $q^2 = 9$ *times).*

Theorem 8 (Chirality and parity, BT869/877)**.** *The matter chirality* $\mathbb{Z}_2$ *is the polar-pair central involution (Steinberg split* $45 + 36$ = *tritangents + double-sixes). Gauge parity is the* W/Q *duality: the four lines through* p_0 *carry* A_4 *inside* PSp *but the full* S_4 *in* $W(E_6)$*; the odd (parity-violating) permutations are exactly the anti-symplectic coset — parity is not an inner symmetry.*

Theorem 9 (Yukawa texture and flavor reflections, BT875/891/893–895)**.** *The cubic coupling on the matter shell is grade-homogeneous in the* R*-eigenbasis:*

$$T_{ijk} = 0 \qquad \textit{unless} \qquad g_i + g_j + g_k \equiv 0 \pmod 3.$$

For a Higgs of grade g_H, the grade-level support on generation pairs is not a pure cyclic shift but the shifted reflection

$$Y_{g_H}[a,b] = 1 \iff b \equiv -a - g_H \pmod 3.$$

The three Higgs-grade skeletons are the three reflections of $D_3 \cong S_3$; their products are generation rotations. Hence the exact $\mathbb{Z}_3$ grade law fixes the allowed Yukawa support and the flavor reflection axes. Since each grade skeleton satisfies

$$Y_{g_H} Y_{g_H}^T = I_3,$$

the grade-level mass operator is triply degenerate and angle-blind. The observable CKM/PMNS angles therefore live in the within-grade $q^2 = 9$ Higgs profile layer, matching the $9 \cdot \mathbf{2}$ standard -doublet multiplicity in the S_3 matter-shell decomposition.

5 Flavor mixing: one scale Φ_3

The texture skeleton above fixes the *structure* of the mixing matrices; their angles are governed by the single substrate scale $\Phi_3 = q^2 + q + 1 = 13$ (the points of $\mathrm{PG}(2, \mathbb{F}_3)$, the Singer C_{13} clock).

Theorem 10 (Unified Φ_3 mixing scale, BT918/920/922)**.** *Both fermion-mixing sectors are deformations at the scale $1/\Phi_3$: the lepton (PMNS) matrix is tribimaximal — the S_3-symmetric point — deformed by $1/\Phi_3$,*

$$\sin^2\theta_{12} = \tfrac{1}{3}\big(1 - \tfrac{1}{\Phi_3}\big) = \tfrac{4}{13}, \quad \sin^2\theta_{23} = \tfrac{1}{2}\big(1 + \tfrac{1}{\Phi_3}\big) = \tfrac{7}{13}, \quad \sin^2\theta_{13} = \tfrac{\lambda}{\Phi_6\Phi_3} = \tfrac{2}{91},$$

while the quark (CKM) Cabibbo angle is the q/Φ_3 rotation off the identity, $\tan\theta_C = q/\Phi_3 = 3/13$. The quark/lepton dichotomy is one scale, two regimes: a small rotation off the identity (quarks) versus a deformation off tribimaximal (leptons). All nine flavor observables (three CKM elements, three PMNS angles, the CP phase, the Jarlskog $J = 27/884000$, and the Wolfenstein $A = 4/5$) match PDG within 1σ except A at 1.7σ; the two large PMNS angles sum to $4/13 + 7/13 = 11/13$ (numerator $11 = k - 1$, the Ihara prime).

6 Dynamics and gravity

Theorem 11 (Gauge connection and curvature, BT882/883/884)**.** *The connection between local gauge groups is flat along collinear edges ($\langle R_p, R_{p'}\rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$) and curved across non-collinear (matter) pairs ($SL(2,3) = 2T$, the 24-cell group). The curvature $F(p,q) = [R_p, R_{p'}]$ is an order-4 quaternion unit of $2T$ on every matter pair ($F^2 = -I$) — an $\mathfrak{su}(2)$-valued field strength supported exactly on the matter graph Q; Wilson loops are abelian ($\mathbb{Z}_3$) on lines and non-abelian (up to $k = 12$) on matter.*

Theorem 12 (Spanning-tree gravity and the Ihara zeta, BT870/872)**.** *The discrete Matrix-Tree gravitational partition function is $\tau(W(3,3)) = 10^{24}16^{15}/40 = 2^{81} \cdot 5^{23}$ — the matter dimension $q^4 = 81$ in the exponent of 2, the tier prime $F_5 = 5$ as base; the complement Q gives $2^{66}3^{39}5^{23}$ with the gauge dimension $q\Phi_3 = 39$. This is the Perron-point ($u = 1$) behavior of the Ihara zeta $\zeta^{-1}(u) = (1-u^2)^{200}(1-u)(1-11u)(1-2u+11u^2)^{24}(1+4u+11u^2)^{15}$, whose nontrivial zeros lie on $|u| = 1/\sqrt{11}$ (graph RH): transport and gravity are one zeta.*

Theorem 13 (The finite spectral triple, BT892/921)**.** *The finite Dirac operator $D = d + d^*$ on the $W(3,3)$ 2-complex has $D^2 = \Delta_0 \oplus \Delta_1 \oplus \Delta_2$ with spectrum $\{0^{122}, 4^{240}, 10^{48}, 16^{30}\}$; its harmonic modes are the homology b_0=1, b_1=81 (Steinberg), b_2=40, and its matter eigenvalue is $\mu = 4$ (multiplicity $|E| = 240$). The spectral action's log-determinant is exactly the gravity τ above. This pins the finite/internal input to the Connes spectral action; the curved-4D continuum (Einstein–Hilbert) limit is the open analytic frontier.*

7 The master theorem

Theorem 14 (The Standard-Model spine, BT886/881)**.** *From the pair $(W(3,3),\ R)$ the discrete Standard Model follows with zero free parameters. Each spacetime point carries a copy of the gauge group $C(R_p)$; the 40 points form a single conjugacy class of local gauge groups (homogeneous spacetime), and the 40 long-root $\mathbb{Z}_3$'s generate all of $\mathrm{PSp}(4, \mathbb{F}_3)$. The gauge group, the generations, the flavor symmetry, the Yukawa texture, chirality, parity, charge-conjugation, the fermion content, and the gauge dynamics are the centralizer, center, normalizer, grading, W/Q duality, and commutator structure of one element of $\mathrm{Sp}(4, \mathbb{F}_3)$, replicated over the 40 points of $W(3,3)$.*

Scope. The discrete structure above is exact and machine-verified (witnesses BT858–895; full-group character and permutation computations over the 25920- and 51840-element groups). What is *not* claimed here: the coupling magnitudes, the precise mixing angles, and the masses — these require continuum/RG input and constitute the established numerical layer — and the curved-4D continuum (spectral-action) limit, which remains the open analytic frontier (the substrate supplies the exact finite spectral input, BT892).

Witness ledger. BT858 Heisenberg shell; BT861 Steinberg register; BT862 H_2; BT863 three generations; BT869 chirality; BT874 texture triality; BT875/891 Yukawa selection; BT893 shifted-reflection Yukawa skeleton; BT894 within-grade Higgs profile scan; BT895 flavor reflection/multiplet map; BT876 gauge group; BT877 parity; BT878 charge-conjugation; BT879 flavor S_3; BT880 generations = center; BT881 local gauge groups; BT882/883/884 connection, curvature, flux; BT886 spine; BT887 color/electroweak; BT888 color = matter Heisenberg; BT889 fermion content; BT890 matter cone/E_7; BT892 finite spectral input; BT918/919 mixing-angle falsifiability scorecard; BT920 PMNS = tribimaximal $\pm 1/\Phi_3$; BT921 Hodge–Dirac spectral triple; BT922 unified Φ_3 mixing scale. Each has an executable script in `analysis/` with JSON results in `data/`.